

Scenario: Pizza Delivery Order System

You are building a simple C program for a pizza shop to take orders and calculate the total cost.

Question 1:

Ask the user to choose a pizza size: 1. Small, 2. Medium, 3. Large. Print the chosen size. If they choose wrong, print "Invalid size."

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int size;
6      printf("Enter Pizza Size:\n");
7      printf("1.Small\n2.Medium\n3.Large\n");
8      scanf("%d",&size);
9      switch(size)
10     {
11         case 1:
12             printf("Small");
13             break;
14         case 2:
15             printf("Medium");
16             break;
17         case 3:
18             printf("Large");
19             break;
20         default:
21             printf("Invalid Size");
22     }
23     return 0;
24 }
```

```
Enter Pizza Size:
1.Small
2.Medium
3.Large
1
Small
-----
Process exited after 2.293 seconds with return value 0
Press any key to continue . . .
```

```
Enter Pizza Size:
1.Small
2.Medium
3.Large
2
Medium
-----
Process exited after 2.037 seconds with return value 0
Press any key to continue . . .
```

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```
Enter Pizza Size:
1.Small
2.Medium
3.Large
3
Large
-----
Process exited after 3.588 seconds with return value 0
Press any key to continue . . .
```

```
Enter Pizza Size:
1.Small
2.Medium
3.Large
5
Invalid Size
-----
Process exited after 2.943 seconds with return value 0
Press any key to continue . . .
```

Question 2:

A small pizza costs \$8. Ask how many pizzas they want. If they want more than one, print "Check our multi-pizza deals." Otherwise, print "Your total is \$8."

```
1  #include<stdio.h>
2
3  int main()
4  {
5      int size, number;
6      printf("Enter Pizza Size:\n");
7      printf("1.Small\n2.Medium\n3.Large\n");
8      scanf("%d",&size);
9      switch(size)
10     {
11         case 1:
12             printf("Small");
13             printf("\nA small pizza costs $8. Ask how many pizzas you want.\t");
14             scanf("%d",&number);
15             if(number>1)
16                 printf("Check our multi-pizza deals.");
17             else
18                 printf("Your total is $8.");
19             break;
20         case 2:
21             printf("Medium");
22             break;
23         case 3:
24             printf("Large");
25             break;
26         default:
27             printf("Invalid Size");
28     }
29     return 0;
30 }
```

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```
Enter Pizza Size:
1.Small
2.Medium
3.Large
1
Small
A small pizza costs $8. Ask how many pizzas you want. 1
Your total is $8.
-----
Process exited after 7.217 seconds with return value 0
Press any key to continue . . .
```

```
Enter Pizza Size:
1.Small
2.Medium
3.Large
1
Small
A small pizza costs $8. Ask how many pizzas you want. 3
Check our multi-pizza deals.
-----
Process exited after 4.379 seconds with return value 0
Press any key to continue . . .
```

Question 3:

Ask how many pizzas they want (1, 2, or 3). 1 pizza costs \$8, 2 pizzas cost \$15, and 3 pizzas cost \$21. Calculate and print the total cost with number of pizza.

```
#include<stdio.h>

int main()
{
    int number;
    printf("1 pizza cost: $8 \n2 pizzas cost: $15 \n3 pizzas cost: $21\t");
    scanf("%d",&number);
    if(number==1)
        printf("Total Cost of %d pizza is: $8",number);
    else if(number==2)
        printf("Total Cost of %d pizzas is: $15",number);
    else
        printf("Total cost of %d pizzas is: $21",number);
    return 0;
}
```

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```
1 pizza costs $8
2 pizzas cost $15
3 pizzas cost $21.
How many pizzas you want.(1, 2 or 3):  1
Total Cost of 1 pizza:  $8
-----
Process exited after 6.052 seconds with return value 0
Press any key to continue . . .
```

```
1 pizza costs $8
2 pizzas cost $15
3 pizzas cost $21.
How many pizzas you want.(1, 2 or 3):  2
Total Cost 2 pizzas:    $15
-----
Process exited after 2.018 seconds with return value 0
Press any key to continue . . .
```

```
1 pizza costs $8
2 pizzas cost $15
3 pizzas cost $21.
How many pizzas you want.(1, 2 or 3):  3
Total Cost 3 pizzas:    $21
-----
Process exited after 1.995 seconds with return value
Press any key to continue . . .
```

Question 4:

Ask for crust type and add the cost to the total. Print the new total Cost.

- Regular crust: \$0
- Thin crust: \$1
- Stuffed crust: \$2

```
#include<stdio.h>

int main()
{
    int number, crust, price, cost, total;
    printf("1 pizza cost: $8 \n2 pizzas cost: $15 \n3 pizzas cost: $21\n");
    printf("Enter Number of Pizzas:\t");
    scanf("%d",&number);
    if(number == 1)
        price=8;
    else if(number == 2)
        price=15;
    else if(number == 3)
        price=21;
    else
        printf("Invalid Number of Pizzas");
    printf("1. Regular Crust: $0 \n2. Thin Crust: $1 \n3. Stuffed Crust: $2\n");
    printf("Enter Crust Type:\t");
    scanf("%d",&crust);
    if(crust == 1)
        cost=0;
    else if(crust == 2)
        cost=1;
    else if(crust == 3)
        cost=2;
    else
        printf("Invalid Crust Type");
    total=price+cost;
    printf("Total Cost is: %d",total);
    return 0;
}
```

```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 2
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      3
Total Cost is: 17
-----
Process exited after 13.56 seconds with return value 0
Press any key to continue . . .
```

```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 3
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      1
Total Cost is: 21
-----
Process exited after 11.23 seconds with return value 0
Press any key to continue . . .
```

Question 5:

Ask if they want extra cheese (Y/N). If yes, add \$1.50 to the total.

```
#include<stdio.h>

int main()
{
    int number, crust, price, cost;
    float total;
    char cheese;
    printf("1 pizza cost: $8 \n2 pizzas cost: $15 \n3 pizzas cost: $21\n");
    printf("Enter Number of Pizzas:\t");
    scanf("%d",&number);
    if(number == 1)
        price=8;
    else if(number == 2)
        price=15;
    else if(number == 3)
        price=21;
    else
        printf("\nInvalid Number of Pizzas");
    printf("1. Regular Crust: $0 \n2. Thin Crust: $1 \n3. Stuffed Crust: $2\n");
    printf("Enter Crust Type:\t");
    scanf("%d",&crust);
    if(crust == 1)
        cost=0;
    else if(crust == 2)
        cost=1;
    else if(crust == 3)
        cost=2;
    else
        printf("Invalid Crust Type");
    total=price+cost;
    printf("Do you want extra cheese? (Y/N): ");
    scanf(" %c", &cheese);
    cheese = toupper(cheese);
    if (cheese == 'Y')
        total += 1.50;
    printf("Total Cost is: %.2f",total);
    return 0;
}
```

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```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 3
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      1
Do you want extra cheese? (Y/N): Y
Total Cost is: 22.50
-----
Process exited after 10.61 seconds with return value 0
Press any key to continue . . .
```

```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 1
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      1
Do you want extra cheese? (Y/N): N
Total Cost is: 8.00
-----
Process exited after 16.76 seconds with return value 0
Press any key to continue . . .
```

Question 6:

Between 11 AM and 2 PM, there's a 10% discount. Ask for the current time (0-23). If the time is between 11 and 14, apply the discount. Print the discounted total.

```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 3
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      3
Do you want extra cheese? (Y/N): Y

Enter Current Time(0-23):      12
Total Cost is: 22.05
-----
Process exited after 18.95 seconds with return value 0
Press any key to continue . . .
```

```
#include<stdio.h>

int main()
{
    int number, crust, price, cost;
    float total, time;
    char cheese;
    printf("1 pizza cost: $8 \n2 pizzas cost: $15 \n3 pizzas cost: $21\n");
    printf("Enter Number of Pizzas:\t");
    scanf("%d",&number);
    if(number == 1)
        price=8;
    else if(number == 2)
        price=15;
    else if(number == 3)
        price=21;
    else
        printf("\nInvalid Number of Pizzas");
    printf("1. Regular Crust: $0 \n2. Thin Crust: $1 \n3. Stuffed Crust: $2\n");
    printf("Enter Crust Type:\t");
    scanf("%d",&crust);
    if(crust == 1)
        cost=0;
    else if(crust == 2)
        cost=1;
    else if(crust == 3)
        cost=2;
    else
        printf("Invalid Crust Type");
    total=price+cost;
    printf("Do you want extra cheese? (Y/N): ");
    scanf(" %c", &cheese);
    cheese = toupper(cheese);
    if (cheese == 'Y')
        total += 1.50;
    printf("\nEnter Current Time (0-23):\t");
    scanf("%f",&time);
    if(time>=11 || time<=14)
        total-= (total*0.10);
    printf("Total Cost is: %.2f",total);
    return 0;
}
```

Question 7:

Students get \$2 off. Ask if they have a student ID (Y/N). If yes, subtract \$2 from the total. Print the final cost.


```
#include<stdio.h>

int main()
{
    int number, crust, price, cost, mode;
    float total, time;
    char cheese, student;
    printf("1 pizza cost: $8 \n2 pizzas cost: $15 \n3 pizzas cost: $21\n");
    printf("Enter Number of Pizzas:\t");
    scanf("%d",&number);
    if(number == 1)
        price=8;
    else if(number == 2)
        price=15;
    else if(number == 3)
        price=21;
    else
        printf("\nInvalid Number of Pizzas");
    printf("1. Regular Crust: $0 \n2. Thin Crust: $1 \n3. Stuffed Crust: $2\n");
    printf("Enter Crust Type:\t");
    scanf("%d",&crust);
    if(crust == 1)
        cost=0;
    else if(crust == 2)
        cost=1;
    else if(crust == 3)
        cost=2;
    else
        printf("Invalid Crust Type");
    total=price+cost;
    printf("Do you want extra cheese? (Y/N): ");
    scanf(" %c", &cheese);
    cheese = toupper(cheese);
    if (cheese == 'Y')
        total += 1.50;
    printf("\nEnter Current Time(0-23):\t");
    scanf("%f",&time);
    if(time>=11 || time<=14)
        total-=(total*0.10);
    printf("Do You Have Student ID (Y/N):\t");
    scanf(" %c",&student);
    student = toupper(student);
    if(student == 'Y')
        total-=2;
    printf("\nTotal Cost is: %.2f",total);
    return 0;
}
```

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```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 3
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      3
Do you want extra cheese? (Y/N): Y

Enter Current Time(0-23):      12
Do You Have Student ID (Y/N):  Y
Total Cost is: 20.05
-----
Process exited after 20.49 seconds with return value 0
Press any key to continue . . .
```

Question 8:

If they order 3 pizzas with stuffed crust, print "You get free garlic bread!"

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```
#include<stdio.h>

int main()
{
    int number, crust, price, cost;
    float total, time;
    char cheese, id;
    printf("1 pizza cost: $8 \n2 pizzas cost: $15 \n3 pizzas cost: $21\n");
    printf("Enter Number of Pizzas:\t");
    scanf("%d",&number);
    if(number == 1)
        price=8;
    else if(number == 2)
        price=15;
    else if(number == 3)
        price=21;
    else
        printf("\nInvalid Number of Pizzas");
    printf("1. Regular Crust: $0 \n2. Thin Crust: $1 \n3. Stuffed Crust: $2\n");
    printf("Enter Crust Type:\t");
    scanf("%d",&crust);
    if(crust == 1)
        cost=0;
    else if(crust == 2)
        cost=1;
    else if(crust == 3)
        cost=2;
    else
        printf("Invalid Crust Type");
    total=price+cost;
    printf("Do you want extra cheese? (Y/N): ");
    scanf(" %c", &cheese);
    cheese = toupper(cheese);
    if (cheese == 'Y')
        total += 1.50;
    printf("\nEnter Current Time(0-23):\t");
    scanf("%f",&time);
    if(time>=11 || time<=14)
        total-=(total*0.10);
    printf("Do You Have Student ID (Y/N):\t");
    scanf(" %c",&id);
    cheese = toupper(cheese);
    if(id == 'Y')
        total-=2;
    if(number==3 && crust==3)
        printf("You get free garlic bread!");
    printf("Total Cost is: %.2f",total);
    return 0;
}
```

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```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 3
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type: 3
Do you want extra cheese? (Y/N): Y

Enter Current Time(0-23): 16
Do You Have Student ID (Y/N): N
You get free garlic bread!
Total Cost is: 22.05
-----
Process exited after 11.92 seconds with return value 0
Press any key to continue . . .
```

Question 9:

If they want delivery, add \$3. Ask if it's delivery or pickup (1. Pickup, 2. Delivery). Calculate the final cost.

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```
#include<stdio.h>

int main()
{
    int number, crust, price, cost, mode;
    float total, time;
    char cheese, id;
    printf("1 pizza cost: $8 \n2 pizzas cost: $15 \n3 pizzas cost: $21\n");
    printf("Enter Number of Pizzas:\t");
    scanf("%d",&number);
    if(number == 1)
        price=8;
    else if(number == 2)
        price=15;
    else if(number == 3)
        price=21;
    else
        printf("\nInvalid Number of Pizzas");
    printf("1. Regular Crust: $0 \n2. Thin Crust: $1 \n3. Stuffed Crust: $2\n");
    printf("Enter Crust Type:\t");
    scanf("%d",&crust);
    if(crust == 1)
        cost=0;
    else if(crust == 2)
        cost=1;
    else if(crust == 3)
        cost=2;
    else
        printf("Invalid Crust Type");
    total=price+cost;
    printf("Do you want extra cheese? (Y/N): ");
    scanf(" %c", &cheese);
    cheese = toupper(cheese);
    if (cheese == 'Y')
        total += 1.50;
    printf("\nEnter Current Time(0-23):\t");
    scanf("%f",&time);
    if(time>=11 || time<=14)
        total-=(total*0.10);
    printf("Do You Have Student ID (Y/N):\t");
    scanf(" %c",&id);
    cheese = toupper(cheese);
    if(id == 'Y')
        total-=2;
    if(number==3 && crust==3)
        printf("You get free garlic bread!\n");
    printf("1.Pickup \n2.Delivery:\t");
    scanf("%d",&mode);

    if(mode==2)
        total+=3;
    printf("\nTotal Cost is: %.2f",total);
    return 0;
}
```

```
1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 3
1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      3
Do you want extra cheese? (Y/N): Y

Enter Current Time(0-23):      16
Do You Have Student ID (Y/N):  N
You get free garlic bread!
1.Pickup
2.Delivery:      2

Total Cost is: 25.05
-----
Process exited after 16.46 seconds with return value 0
Press any key to continue . . .
```

Question 10:

Combine all into one program:

- Ask for size, number of pizzas, crust type, extra cheese, time, student status, and delivery option.
- Calculate the total cost step by step.
- Print a receipt with the final amount and any free items.

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```
#include<stdio.h>

int main()
{
    int size, number, crust, price, cost, mode;
    float total, time;
    char cheese, id;
    printf("1.Small\n2.Medium\n3.Large\n");
    printf("Enter Pizza Size:\t");
    scanf("%d",&size);
    printf("\n1.pizza cost: $8 \n2.pizzas cost: $15 \n3.pizzas cost: $21\n");
    printf("Enter Number of Pizzas:\t");
    scanf("%d",&number);
    if(number == 1)
        price=8;
    else if(number == 2)
        price=15;
    else if(number == 3)
        price=21;
    else
        printf("\nInvalid Number of Pizzas");
    printf("\n1. Regular Crust: $0 \n2. Thin Crust: $1 \n3. Stuffed Crust: $2\n");
    printf("Enter Crust Type:\t");
    scanf("%d",&crust);
    if(crust == 1)
        cost=0;
    else if(crust == 2)
        cost=1;
    else if(crust == 3)
        cost=2;
    else
        printf("Invalid Crust Type");
    total=price+cost;
    printf("\nDo you want extra cheese? (Y/N): ");
    scanf(" %c", &cheese);
    cheese = toupper(cheese);
    if (cheese == 'Y')
        total += 1.50;
    printf("\nEnter Current Time(0-23):\t");
    scanf("%f",&time);
    if(time>=11 && time<=14)
        total-=(total*0.10);
    printf("\nDo You Have Student ID (Y/N):\t");
    scanf(" %c",&id);
    cheese = toupper(cheese);
    if(id == 'Y')
        total-=2;
    printf("\n1.Pickup \n2.Delivery:\t");
    scanf("%d",&mode);

    if(mode==2)
        total+=3;
    if(number==3 && crust==3)
        printf("\n\nYou get free garlic bread!\n");
    printf("\nTotal Cost is: %.2f",total);
    return 0;
}
```

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```
1.Small
2.Medium
3.Large
Enter Pizza Size:      2

1 pizza cost: $8
2 pizzas cost: $15
3 pizzas cost: $21
Enter Number of Pizzas: 3

1. Regular Crust: $0
2. Thin Crust: $1
3. Stuffed Crust: $2
Enter Crust Type:      3

Do you want extra cheese? (Y/N): N

Enter Current Time(0-23):      20

Do You Have Student ID (Y/N):  Y

1.Pickup
2.Delivery:      2

You get free garlic bread!

Total Cost is: 24.00
-----
Process exited after 16.78 seconds with return value 0
```